Lecture 14

Asymptotics of integrals

Objectives. *A large parameter can concentrate an integral near an endpoint or a stationary point of its phase. We prove Watson's lemma for a decaying exponential and compute the Fresnel integral by contour rotation. Stationary phase then gives oscillatory leading terms. The Airy integral shows how contour deformation turns an oscillatory integral into a Gaussian approximation.*

By the end of this lecture you will be able to:

- Interpret an asymptotic expansion without assuming its convergence.
- Prove and apply Watson's lemma under an exponential growth bound.
- Identify interior, endpoint and double-zero stationary-phase terms.
- Derive the Airy equation from an absolutely convergent contour integral.

14.1 Asymptotic expansions

The large parameter in this lecture is $x \to +\infty$. For complex functions, $f = O(g)$ means $|f(x)| \le C|g(x)|$ for all sufficiently large x, with a constant C. The notation $f = o(g)$ means $f(x)/g(x) \to 0$, and $f \sim g$ means $f(x)/g(x) \to 1$. Denominators are assumed nonzero for all sufficiently large x.

Definition 14.1. An *asymptotic scale* is a sequence of functions $(\phi_n)_{n\ge 0}$ with $\phi_{n+1} = o(\phi_n)$. An *asymptotic expansion* in this scale, written $f \sim \sum_{n\ge 0} a_n \phi_n$, means that for every fixed $N \ge 1$,

$$f(x) - \sum_{n=0}^{N-1} a_n \phi_n(x) = o(\phi_{N-1}(x)).$$

The limit concerns x, with N fixed. It makes no assertion about the infinite series at a fixed x. For example, the Stieltjes integral

$$J(x) = \int_0^\infty \frac{e^{-xt}}{1+t}\,\mathrm{d}t$$

has the expansion $\sum_{n\ge 0}(-1)^n n! x^{-n-1}$, as Watson's lemma below shows. At any fixed $x > 0$, the ratio of successive term magnitudes is $(n+1)/x$, so the series diverges. Its finite truncations still approximate $J(x)$ as $x \to \infty$.

14.2 Watson's lemma

The exponential e^{-xt} suppresses every fixed region $t \geq \delta > 0$ when x is large. Only the expansion of the remaining factor near zero contributes powers of $1/x$. The source convention is Ablowitz–Fokas, §6.2.2.[1]

Theorem 14.2 (Watson). *Let $\beta > -1$. Suppose $g : [0,\infty) \to \mathbb{C}$ is measurable, smooth on $[0,\delta_0]$ for some $\delta_0 > 0$, and satisfies $|g(t)| \leq Ce^{bt}$ for $t \geq 0$, with constants $C > 0$ and $b \in \mathbb{R}$. Then, as $x \to +\infty$,*

$$\int_0^\infty t^\beta g(t) e^{-xt}\,\mathrm{d}t \sim \sum_{n\geq 0} \frac{g^{(n)}(0)\Gamma(n+\beta+1)}{n!\,x^{n+\beta+1}}.$$

After N terms the remainder is $O(x^{-N-\beta-1})$.

Proof. The integral is absolutely convergent for $x > \max(b, 0)$. Fix $N \geq 1$ and $0 < \delta < \delta_0$. Taylor's theorem gives

$$g(t) = P_{N-1}(t) + R_N(t), \qquad P_{N-1}(t) = \sum_{n=0}^{N-1} \frac{g^{(n)}(0)}{n!} t^n, \qquad |R_N(t)| \leq C_N t^N \quad (0 \leq t \leq \delta).$$

On this interval the remainder contributes at most

$$C_N \int_0^\infty t^{N+\beta} e^{-xt}\,\mathrm{d}t = C_N \Gamma(N+\beta+1) x^{-N-\beta-1}.$$

The substitution $u = xt$ gives the equality from the gamma integral of (13.1).

The tails of g and the polynomial are exponentially small. For x sufficiently large, $x - b \geq x/2 + 1$, so

$$\int_\delta^\infty t^\beta |g(t)| e^{-xt}\,\mathrm{d}t \leq Ce^{-x\delta/2} \int_\delta^\infty t^\beta e^{-t}\,\mathrm{d}t.$$

Each polynomial monomial has the same kind of bound. We may therefore replace the integrals of P_{N-1} over $[0,\delta]$ by integrals over $[0,\infty)$ at exponentially small cost. For each monomial,

$$\int_0^\infty t^{n+\beta} e^{-xt}\,\mathrm{d}t = \Gamma(n+\beta+1) x^{-n-\beta-1}.$$

Combining these estimates proves the remainder claim, which is stronger than the defining little-o remainder of the asymptotic expansion. ∎

For the Stieltjes integral, take $\beta = 0$ and $g(t) = (1+t)^{-1}$. Its derivatives satisfy $g^{(n)}(0) = (-1)^n n!$, so Watson gives

$$J(x) = \frac{1}{x} - \frac{1}{x^2} + \frac{2}{x^3} + O(x^{-4}).$$

The first omitted order is an assertion as $x \to \infty$, rather than a convergence test for the infinite series.

14.3 Fresnel integrals and stationary phase

A *stationary point* of a real smooth phase $\varphi(t)$ is a point t_0 with $\varphi'(t_0) = 0$. It is *nondegenerate* if $\varphi''(t_0) \neq 0$. Quadratic oscillations near such a point give a Fresnel integral.

[1] Their large parameter k is written x here. Their endpoint-power expansion specialises to $t^\beta g(t)$.

Proposition 14.3 (Fresnel). *As an improper integral,*

$$\int_0^\infty e^{\mathrm{i}t^2}\,\mathrm{d}t = \frac{\sqrt{\pi}}{2}e^{\mathrm{i}\pi/4}.$$

Proof. Integrate e^{-z^2} around the sector bounded by $[0,R]$, the circular arc from R to $Re^{\mathrm{i}\pi/4}$ and the ray back to zero. The integrand is entire, so Cauchy's theorem on $\mathbb{C}$ (Theorem 9.4) gives zero around the boundary. On the arc, put $u = \pi/4 - \arg z$. Since $\cos(2\arg z) = \sin(2u) \geq 4u/\pi$, its integral has absolute value at most

$$R\int_0^{\pi/4} e^{-4R^2u/\pi}\,\mathrm{d}u \leq \frac{\pi}{4R}.$$

The oriented ray integral is $-e^{\mathrm{i}\pi/4}\int_0^R e^{-\mathrm{i}t^2}\,\mathrm{d}t$. Letting $R\to\infty$ and using the real Gaussian integral (Theorem 4.3) gives

$$\frac{\sqrt{\pi}}{2} = e^{\mathrm{i}\pi/4}\int_0^\infty e^{-\mathrm{i}t^2}\,\mathrm{d}t.$$

This also proves existence of the improper integral. Complex conjugation gives the stated formula. The convergence is conditional, since the integrand has absolute value one. ∎

Theorem 14.4 (Stationary phase · *sketched*). *Let $a\in C_c^\infty((A,B))$ and let the phase $\varphi(t)$ be real and smooth on $[A,B]$. Suppose its only stationary point is $t_0\in(A,B)$, with $\varphi''(t_0)\neq 0$ and $a(t_0)\neq 0$. Then*

$$\begin{aligned}\int_A^B a(t)e^{\mathrm{i}x\varphi(t)}\,\mathrm{d}t &\sim a(t_0)\sqrt{\frac{2\pi}{x|\varphi''(t_0)|}}\\ &\quad\times e^{\mathrm{i}x\varphi(t_0)+\mathrm{i}\pi\operatorname{sgn}(\varphi''(t_0))/4}.\end{aligned}$$

If $t_0 = A$, replace the interior-location and support assumptions by the following: assume $a(A)\neq 0$, $\varphi'(A)=0$, $\varphi''(A)\neq 0$, that a is smooth on $[A,B]$ and vanishes near B, and that there are no other stationary points. The leading term is half this expression.

Proof sketch. Away from t_0, use $e^{\mathrm{i}x\varphi(t)} = (\mathrm{i}x\varphi'(t))^{-1}(e^{\mathrm{i}x\varphi(t)})'$. Repeated integration by parts after a smooth cutoff makes that contribution $O(x^{-M})$ for every fixed M. Near t_0, there is a smooth coordinate u with

$$\varphi(t) = \varphi(t_0) + \frac{\sigma}{2}u^2, \qquad u(t_0)=0, \qquad u'(t_0) = \sqrt{|\varphi''(t_0)|}, \qquad \sigma = \operatorname{sgn}(\varphi''(t_0)).$$

Indeed, take the signed square root of $2\sigma(\varphi(t)-\varphi(t_0))$; Taylor's theorem makes its quotient by $t-t_0$ smooth and positive. The transformed amplitude has value $a(t_0)/\sqrt{|\varphi''(t_0)|}$ at zero. Rescaling $v=\sqrt{x}\,u$ gives the leading Fresnel factor $\sqrt{2\pi}e^{\mathrm{i}\sigma\pi/4}x^{-1/2}$ on the full line. Integration by parts bounds the rescaled tails outside $|v|\leq R$ by C/R, uniformly for large x, and then $R\to\infty$ removes them. At an endpoint only one half of the quadratic neighbourhood occurs. The localisation and remainder details are sketched here. ∎

For example, if $a\in C_c^\infty(\mathbb{R})$ and $a(0)=2$, the phase $\varphi(t)=t^2/2$ gives $\int a(t)e^{\mathrm{i}xt^2/2}\,\mathrm{d}t \sim 2\sqrt{2\pi/x}\,e^{\mathrm{i}\pi/4}$. A nonstationary endpoint with nonzero amplitude instead contributes $O(x^{-1})$, from the integration-by-parts boundary term.

A vanishing amplitude changes the order. Keep the interior phase and compact-support hypotheses, but now suppose $\varphi''(t_0)>0$ and $a(t)=(t-t_0)^2b(t)$, where b is smooth

near t_0 and $b(t_0) \neq 0$. In the quadratic coordinate the local amplitude is $u^2 b_1(u)$, where $b_1(0) = b(t_0)/\varphi''(t_0)^{3/2}$. With a compact cutoff, integration by parts gives

$$\int u^2 b_1(u) e^{\mathrm{i}xu^2/2}\,\mathrm{d}u = \frac{\mathrm{i}}{x}\int (b_1(u) + u b_1'(u)) e^{\mathrm{i}xu^2/2}\,\mathrm{d}u.$$

Applying the same stationary-phase leading term to the right side gives

$$\int_A^B a(t) e^{\mathrm{i}x\varphi(t)}\,\mathrm{d}t \sim b(t_0)\sqrt{2\pi}\, e^{3\pi\mathrm{i}/4}(x\varphi''(t_0))^{-3/2} e^{\mathrm{i}x\varphi(t_0)}. \tag{14.1}$$

This remains a sketched consequence of stationary phase. The compact-support condition removes nonstationary endpoint terms of order x^{-1}, which could otherwise dominate the $x^{-3/2}$ term. No ordinary improper integral of $u^2 e^{\mathrm{i}u^2/2}$ is asserted.

14.4 The Airy contour and steepest descent

For a complex argument z, define the Airy function by the contour integral

$$\operatorname{Ai}(z) = \frac{1}{2\pi\mathrm{i}}\int_C e^{t^3/3 - zt}\,\mathrm{d}t. \tag{14.2}$$

The contour C comes from infinity on the ray $\arg t = -\pi/3$, runs to zero, and leaves along the ray $\arg t = \pi/3$. On either ray, the cubic term is $-r^3/3$, whereas $|zt| \le |z|r$. The integral therefore converges absolutely, uniformly on compact sets of z. Every derivative in z adds only a power of t and has the same integrable bound. The holomorphic parameter theorem (Theorem 10.9) therefore makes Ai entire.

Proposition 14.5. *The contour integral satisfies* $\operatorname{Ai}''(z) = z\operatorname{Ai}(z)$. *For* $x \in \mathbb{R}$ *it agrees with the improper oscillatory integral*

$$\operatorname{Ai}(x) = \frac{1}{\pi}\int_0^\infty \cos(t^3/3 + xt)\,\mathrm{d}t.$$

Proof. Differentiating the absolutely convergent integral twice gives

$$\operatorname{Ai}''(z) - z\operatorname{Ai}(z) = \frac{1}{2\pi\mathrm{i}}\int_C (t^2 - z) e^{t^3/3 - zt}\,\mathrm{d}t = \frac{1}{2\pi\mathrm{i}}\int_C \frac{\mathrm{d}}{\mathrm{d}t} e^{t^3/3 - zt}\,\mathrm{d}t = 0.$$

The endpoint values vanish by cubic decay. There is no boundary term at zero, since the two contour pieces join there.

For real x, rotate the upper ray to the positive imaginary axis, and the lower ray to the negative imaginary axis. On the upper connecting arc write $\arg t = \pi/2 - \delta$, $0 \le \delta \le \pi/6$. Its exponent has real part at most

$$-\frac{2R^3}{\pi}\delta + |x|R\delta.$$

Here $\sin(3\delta) \ge 6\delta/\pi$ and $|\cos(\arg t)| \le \delta$. Multiplying its exponential by the arc length element $R\,\mathrm{d}\delta$ gives a bound $O(R^{-2})$ for fixed x. The lower arc has the same bound. Cauchy's theorem therefore transfers the integral to the oriented imaginary axis. Substitute $t = \mathrm{i}u$ to obtain

$$\operatorname{Ai}(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\mathrm{i}(u^3/3 + xu)}\,\mathrm{d}u.$$

The symmetric contributions are conjugates and give the cosine formula. The one-sided improper limits also exist: integration by parts with the phase derivative $u^2 + x$ gives tails $O(R^{-2})$ once $R^2 > 2|x|$. Changing u to $-u$ gives the equivalent positive-sign full-line formula. This agrees with Stein–Shakarchi, Appendix A, §3.[2] ∎

A *nondegenerate saddle* of a holomorphic phase p satisfies $p' = 0$ and $p'' \neq 0$. A *constant-phase path* has constant imaginary part of p. Along a path of *steepest descent*, that imaginary part is constant and the real part decreases away from the saddle. Beals–Wong discuss this geometry in §13.5.[3] For $x > 0$, set $t = \sqrt{x}\,w$. The phase becomes $x^{3/2}p(w)$ with $p(w) = w^3/3 - w$ and saddles $w = 1$ and $w = -1$. The contour through 1 has the same end directions as C. Writing $w = u + \mathrm{i}v$, its exact constant-phase equation is

$$u = \sqrt{1 + v^2/3}, \qquad \operatorname{Im} p = 0, \qquad \operatorname{Re} p = -\frac{2}{3}u - \frac{8}{9}uv^2.$$

The real part has its unique maximum $-2/3$ at $w = 1$ and decreases towards both ends. Its local expansion is $-2/3 - v^2 + O(v^4)$.

The contour geometry is shown together with the Fresnel rotation.

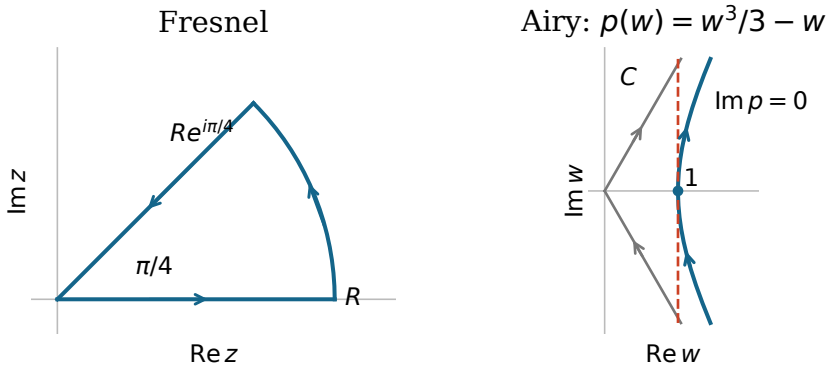


Figure 14.1: Oriented contours. The Fresnel sector rotates a Gaussian to a quadratic oscillation. For Airy, the grey rays form C in the scaled plane. The curved solid path is the constant-phase path through 1; the dashed line has the same tangent there.

Proposition 14.6 (Positive Airy asymptotic · *sketched*). *As $x \to +\infty$,*

$$\operatorname{Ai}(x) \sim \frac{e^{-2x^{3/2}/3}}{2\sqrt{\pi}\,x^{1/4}}.$$

Proof sketch. The deformation to the constant-phase path is sketched here. The constant-phase path gives a Gaussian near the contributing saddle. One can compute the leading term using the parallel vertical line $t = \sqrt{x} + \mathrm{i}u$. Shifting the imaginary-axis contour to this line is allowed: on a horizontal connector $t = v \pm \mathrm{i}R$ with $0 \le v \le \sqrt{x}$, the real part of the exponent is $v^3/3 - vR^2 - xv \le -vR^2$. Its integral is at most R^{-2}. Thus the exact shifted-line

[2] Their real Airy argument s is x here. Substituting $t = -\mathrm{i}u$ in their positive-sign oscillatory formula gives our lower-to-upper ray orientation.

[3] Their large parameter λ corresponds to x in a general asymptotic integral here. For the scaled Airy phase, the multiplier is $x^{3/2}$.

formula is

$$\operatorname{Ai}(x) = \frac{e^{-2x^{3/2}/3}}{2\pi} \int_{\mathbb{R}} e^{-\sqrt{x}u^2} e^{-\mathrm{i}u^3/3}\, \mathrm{d}u.$$

Set $u = x^{-1/4}v$. The remaining integral tends to $\int_{\mathbb{R}} e^{-v^2}\, \mathrm{d}v = \sqrt{\pi}$ by dominated convergence (Theorem 3.5) and Theorem 4.3, since the cubic factor has absolute value one. The Jacobian $x^{-1/4}$ and the prefactor $1/(2\pi)$ give the stated constant. ∎

The other real regime and the associated turning-point approximations are developed in Lecture 28.

14.5 The pendulum near its limiting amplitude

A *libration* is a back-and-forth oscillation about the lower pendulum equilibrium, rather than a full rotation. Its amplitude is the largest angular displacement from that equilibrium. The modulus k is the sine of half this amplitude, with $0 < k < 1$. Its *complementary modulus* is $k' = \sqrt{1 - k^2}$. This prime does not denote a derivative. The pendulum period in Theoretical Mechanics is $4K(k)$ in units where $\nu_0 = 1$. Rescaling time by $\tau = \nu_0 t$ gives $4K(k)/\nu_0$ for the pendulum of Lecture 2, where

$$K(k) = \int_0^{\pi/2} \frac{\mathrm{d}u}{\sqrt{1 - k^2 \sin^2 u}}.$$

We use this mechanical period relation as an input here; its derivation from the pendulum equation is given in Lecture 21. The present argument estimates the elliptic integral in that relation. As $k \to 1$, the amplitude approaches the upper equilibrium angle. The period diverges logarithmically.

Proposition 14.7 (Elliptic-integral limit · *sketched*). *As $k \to 1$ from below, $K(k) = \ln(4/k') + o(1)$.*

Proof sketch. The matching argument is given below; its logarithmic error bound is sketched. Put $\delta = k'$ and substitute $t = \cos u$:

$$K(k) = \int_0^1 \frac{\mathrm{d}t}{\sqrt{1 - t^2}\sqrt{\delta^2 + (1 - \delta^2)t^2}}.$$

Replacing the second square root by $\sqrt{\delta^2 + t^2}$ changes this by $o(1)$. Indeed, rationalising the reciprocal difference bounds it by $C\delta^2/\sqrt{\delta^2 + t^2}$; after multiplication by $(1 - t^2)^{-1/2}$ its integral is $O(\delta^2 \ln(1/\delta))$. Subtract the explicitly integrable singular part. Dominated convergence (Theorem 3.5) gives

$$K(k) - \int_0^1 \frac{\mathrm{d}t}{\sqrt{\delta^2 + t^2}} \longrightarrow \int_0^1 \frac{(1 - t^2)^{-1/2} - 1}{t}\, \mathrm{d}t = \ln 2.$$

For the domination use $\sqrt{\delta^2 + t^2} \geq t$; the displayed limiting integrand is integrable at both endpoints. Its value follows from $t = \sin u$, which turns it into $\int_0^{\pi/2} \tan(u/2)\, \mathrm{d}u = \ln 2$. The subtracted integral is $\ln((1 + \sqrt{1 + \delta^2})/\delta) = \ln(2/\delta) + o(1)$. Adding the constants gives $\ln(4/\delta) + o(1)$. ∎

The libration period grows without bound as the amplitude tends to the upper equilibrium angle. The logarithm gives the rate at which this family of periods diverges. The constant agrees with NIST DLMF, equation 19.12.1.

Summary.

- An asymptotic expansion controls every fixed finite truncation.
- For $\beta > -1$, Watson's lemma uses smoothness near zero and an exponential growth bound to extract the endpoint coefficients.
- With the stated smoothness and support assumptions, a nondegenerate stationary point with nonzero amplitude gives order $x^{-1/2}$. A double-zero amplitude gives order $x^{-3/2}$ when endpoint terms are absent.
- The Airy contour gives its differential equation and its exponentially small positive-argument asymptotic. The libration period diverges logarithmically as the modulus tends to one from below.

References

M. J. Ablowitz and A. S. Fokas, *Complex Variables: Introduction and Applications*, second edition, Cambridge, 2003, §§6.2–6.3, pp. 426–429 and 443–446. E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton, 2003, Appendix A, §3, pp. 328–331. R. Beals and R. Wong, *Special Functions and Orthogonal Polynomials*, Cambridge, 2016, §13.5, pp. 342–345.